NRC·CNRC

NRC Publications Archive
Archives des publications du CNRC

Accepting alternate solutions under objective-based codes

Archer, J. W.

This publication could be one of several versions: author's original, accepted manuscript or the publisher's version. / La version de cette publication peut être l'une des suivantes : la version prépublication de l'auteur, la version acceptée du manuscrit ou la version de l'éditeur.

Publisher's version / Version de l'éditeur:

Ontario Building Officials Association Journal, March, pp. 12-13, 2001-03-01

NRC Publications Record / Notice d'Archives des publications de CNRC:
https://nrc-publications.canada.ca/eng/view/object/?id=e5e9e318-6e0f-4fd8-b2a4-0a7d502168bc
https://publications-cnrc.canada.ca/fra/voir/objet/?id=e5e9e318-6e0f-4fd8-b2a4-0a7d502168bc

Access and use of this website and the material on it are subject to the Terms and Conditions set forth at
https://nrc-publications.canada.ca/eng/copyright
READ THESE TERMS AND CONDITIONS CAREFULLY BEFORE USING THIS WEBSITE.

L'accès à ce site Web et l'utilisation de son contenu sont assujettis aux conditions présentées dans le site
https://publications-cnrc.canada.ca/fra/droits
LISEZ CES CONDITIONS ATTENTIVEMENT AVANT D'UTILISER CE SITE WEB.

Questions? Contact the NRC Publications Archive team at PublicationsArchive-ArchivesPublications@nrc-cnrc.gc.ca. If you wish to email the authors directly, please see the first page of the publication for their contact information.

Vous avez des questions? Nous pouvons vous aider. Pour communiquer directement avec un auteur, consultez la première page de la revue dans laquelle son article a été publié afin de trouver ses coordonnées. Si vous n'arrivez pas à les repérer, communiquez avec nous à PublicationsArchive-ArchivesPublications@nrc-cnrc.gc.ca.

National Research Council Canada | Conseil national de recherches Canada

Canada

National Research Council Canada
Conseil national de recherches Canada

NRC · CNRC

Accepting alternate solutions under objective-based codes

Archer, J.W.

NRCC-45014

A version of this paper is published in / Une version de ce document se trouve dans :
Ontario Building Officials Association Journal, March 2001, pp. 12-13

www.nrc.ca/irc/ircpubs

construction
Institute for Research in Construction
Institut de recherche en construction
IRC

Accepting Alternate Solutions under Objective-Based Codes

By John W. Archer
Secretary
Canadian Commission on Building and Fire Codes

Some code users have expressed concern that objective-based codes will bring increased pressure to accept materials, products and designs for buildings that do not conform to the code. Other countries that have reformed their national codes have not seen a significant increase in this pressure and the same outcome can be expected in Canada. However, let's look at the situation.

Building designs, products and construction approaches are currently accepted or approved by authorities using a variety of methods. An experienced building official likely uses both informal and formal approaches to evaluating compliance with the code. These are unlikely to change with the introduction of objective-based codes.

For example, what does the building official do now when a product or piece of equipment is installed in a building? Often a visual inspection is not sufficient to determine whether or not the product meets the code. How do you tell if a fan will provide a required ventilation rate? In these cases the building official relies on information provided by the manufacturer that the material or product complies with a standard. For some particularly critical building items, such as fire doors, the building officials will actually look for labels from third party certification organizations attached to the product. This approach is unlikely to change under objective-based codes.

What happens when there is no standard that the manufacturer of an innovative product or equipment can comply with? In this case, the building code's equivalency provisions are invoked. Here the building official can be presented with some real challenges in determining whether or not a design, material or product will meet the intent of the code. In these instances it is important to remember that the onus on demonstrating compliance rests with the proponent, not the building official and that help is available.

In these cases the building official can request that the person or company proposing the use of the innovative product obtain a third-party evaluation from an organization set up for the task. An example is Ontario's Building Materials Evaluation Commission (BMEC) or the Canadian Construction Materials Centre (CCMC). When approached by a client, CCMC will consult experts, develop an evaluation guide that determines what testing or analysis is needed, and then prepare an evaluation report based on the results of this testing or analysis. This evaluation report is essentially an advisory document on which the building official can base his or her decision to accept a product or not. This approach, as well, is unlikely to change under objective-based codes.

In some cases a building official is asked to accept complex designs for the structure of large buildings or for heating, ventilation and air-conditioning systems. The official would not be able to verify compliance with the code without repeating the calculations

that went into the original design. Some large municipalities may have the resources to do this, but not smaller ones. In these cases, the building official would probably insist on seeing a professional's stamp that indicated that an architect or engineer experienced in the design of the structural or mechanical systems had completed the calculations. Some building officials might require "proof" that the professional did have the necessary experience, or might require that the professional sign a statement confirming that the design complied with the code. This approach is also unlikely to change under objective-based codes.

What happens when a design has some unique features that do not comply with the specific provisions of the code? What does the building official do then? In this instance, an approach might be to require that an independent professional become involved and provide a third-party evaluation and certification that the design meets the intent of the code. This approach would also be applicable to objective-based codes, and will be facilitated by the additional information on intents, functional requirements and objectives available with it.

It is important for the building official to remember that the new Canadian objective-based codes will be different from performance-based codes in other countries. In Canada, the alternate solution must be not just "good enough" to meet qualitative objectives and functional requirements identified in the code; it must provide a level of performance equivalent to that provided by the code's acceptable solutions. The functional requirements serve to identify those areas of performance in which this equivalence will be judged.

What happens if a building official is asked to accept a design that incorporates new or unique features whose performance is clearly inferior to the acceptable solutions in the code but where the designer insists that the performance is adequate? Here the building official should most likely say "No." This is an instance where the official is being asked to make a decision as to what is "good enough." These decisions are best left to appeal boards or to the code development system. In other words, if the performance of the innovative approach is truly "good enough," then the code should be changed. But these are difficult decisions and individual building officials should not be expected to shoulder this responsibility.

An historic example of this process is large atriums in buildings. When first proposed, these atriums clearly did not comply with the code and building officials properly refused to accept them. The collective wisdom and judgment of the committees responsible for the codes was brought to bear on the issue and a series of technical changes to the codes were then developed and implemented to identify the circumstance under which the fire safety of atrium buildings could be judged to be "good enough."

By providing significantly more information on what aspects of performance are important, objective-based codes help suppliers, designers and building officials to identify which of the above scenarios applies to a given situation.

Thus, objective-based codes are not likely to make life more difficult for building officials. They should, in fact, make life easier.

John Archer is a member of the Canadian Codes Centre of NRC's Institute for Research in Construction.